

Exponent Laws with Variable BasesIndependent Set 3.T1 **ANSWER KEY**

Name

Date

Period

Six questions, on your own, in silence. Show the step that does the work.

1. Write as one power. $x^6 \cdot x^9$

$$x^{15}$$

2. Write as a single monomial. $(a^5)^4$

$$a^{20}$$

Power of a power: multiply.

3. Write as a single monomial. $(3y^2)^3$

$$27y^6$$

The 3 is raised too.

4. Write as a single monomial. $(2m^4)(5m^3)$

$$10m^7$$

Coefficients multiply; exponents add.

5. Write as a single monomial. $\frac{20x^9}{4x^3}$

$$5x^6$$

Coefficients divide; exponents subtract.

6. Find the value of n . $y^4 \cdot y^n = y^{10}$

$$n = 6$$

$$4 + n = 10.$$